U2210251

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Page 319/ exercise 56

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$$f(x)=\int_0^{\sin x}\sqrt{1+t^2}\,dt$$

$$g(y)=\int_3^{y} f(x)\,dx$$

$$g''\left(\frac{\pi}{6}\right)=?$$

$$g''(y)=f'(y)$$

$$f'(y)=\sqrt{1+\sin^2 y}\,\cos y$$

$$g''(y)=\sqrt{1+\sin^2 y}\,\cos y$$

$$g''\left(\frac{\pi}{6}\right)=\sqrt{1+\sin^2\left(\frac{\pi}{6}\right)}\,\cos\left(\frac{\pi}{6}\right)=$$

$$=\sqrt{\frac{4}{4}+\frac{1}{4}}\cdot\frac{\sqrt{3}}{2}=\frac{\sqrt{15}}{2}\approx 1{,}94$$

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Page 327 / Ex 41

$$\int_{-1}^{2} (x - 2|x|)\,dx =$$

$$= \int_{-1}^{2} x\,dx - \int_{-1}^{2} 2|x|\,dx =$$

$$= \left[\frac{x^2}{2}\right]\Big|_{-1}^{2} - \left[\int_{0}^{2} 2|x|\,dx + \int_{-1}^{0} 2|x|\,dx\right] =$$

$$= \left[\frac{x^2}{2}\right]\Big|_{-1}^{2} - \int_{0}^{2} 2x\,dx + \int_{-1}^{0} 2x\,dx =$$

$$= \left[\frac{x^2}{2}\right]\Big|_{-1}^{2} - \left[x^2\right]\Big|_{0}^{2} + \left[x^2\right]\Big|_{-1}^{0} =$$

$$= \left(2 - \frac{1}{2}\right) - (4) + (-1) = -3,5$$

Page 336/ Exercise 25

Page 336/ Ex 25

$$\int \sqrt{\cot x}\, \csc^2 x \, dx$$

$$u = \sqrt{\cot x}$$

$$du = \frac{1}{2}\sqrt{tg\, x} \cdot \left(-\frac{1}{\sin^2 x}\right) dx$$

$$dv = \frac{1}{\sin^2 x}\, dx$$

$$\int \sqrt{\cot x}\, \frac{1}{\sin^2 x}\, dx$$

$$-\sqrt{\cot x}^{\,3} - \int \frac{\cot x}{2} \cdot \frac{\sqrt{tg\, x}}{\sin^2 x}\, dx$$

$$-\sqrt{\cot x}^{\,3} - \int \left[\frac{\cos x}{\sin x} \cdot \sqrt{\frac{\sin x}{\cos x}} \cdot \frac{1}{2 \cdot \sin^2 x}\right] dx$$

$$-\sqrt{\cot x}^{\,3} - \int \left[\sqrt{\frac{\cos x}{\sin x}} \cdot \frac{1}{2 \cdot \sin^2 x}\right] dx$$

$$-\sqrt{\cot x}^{\,3} - \frac{1}{2} \cdot \int \left[\sqrt{\cot x} \cdot \frac{1}{\sin^2 x}\right] dx$$

$$\int \sqrt{\cot x}\, \csc^2 x \, dx = m$$

$$-\sqrt{\cot x}^{\,3} - \frac{1}{2} m = m$$

$$m = -\frac{2}{3} \cdot \sqrt{\cot x}^{\,3} + C$$

$$\int \sqrt{\cot x}\, \csc^2 x \, dx = -\frac{2}{3} \cdot \sqrt{\cot x}^{\,3} + C$$

Page 349/ Exercise 20

Solution is divided into 2 parts, second is the continuation of the first one

Page 349/Ex 20

(1) $x = y^4$; (2) $y = \sqrt{2-x}$; (3) $y = 0$

$y = \sqrt{2-x}$

$x = 2 - y^2$

$y^4 = 2 - y^2$

$y^4 + y^2 - 2 = 0$

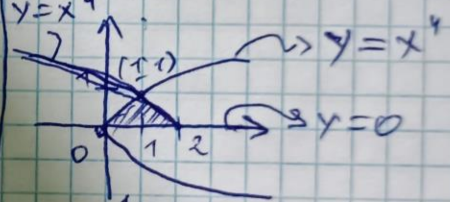
$(y^2+2)(y^2-1) = 0$

$y = \pm 1$ but $y \geq 0$ so $y = 1$

when $y=1$, as $x = y^4$, $x = 1$

~~(1) $x = y^4$; $y(x) = 0 \to x = 0$~~

~~(2) $y(x) = \sqrt{2-x} = 0 \Rightarrow x = 2$~~



$$A = \int_0^1 \left[(2-y^2) - y^4\right] dy = \left[2y - \frac{y^3}{3} - \frac{y^5}{5}\right]\Big|_0^1 =$$

$$= \left(2 - \frac{1}{3} - \frac{1}{5}\right) - 0 = \frac{22}{15} = 1\frac{7}{15} \approx 1{,}47$$